

## PEA307:ADVANCED ANALYTICAL SKILLS-I

L:2 T:1 P:0 Credits:3

**Course Outcomes:** Through this course students should be able to

CO1 :: demonstrate procedural fluency with numerations and mathematical operations to solve the stated problems

CO2 :: use the concepts to solve various percentage and income based questions

CO3 :: analyze the data given and interpret it for solving flow chart and number series

CO4 :: employ the analytical concepts learnt to solve questions of ratio and proportion

CO5 :: use the analytical concepts learnt to solve questions of ratio and proportion

CO6 :: analyze an appropriate approach to solve logical reasoning problems related to blood relations and directions

### Unit I

**Speed maths** : fast calculation techniques

**Advanced numeration** : classification of numbers, divisibility rules, factors, factorials, unit digit and cyclicity, remainder theorems, HCF and LCM

**Mean** : advance methods of mean calculation, combined mean, inclusion and exclusion related problems

### Unit II

**Advanced percentage** : percentage to fraction conversion, successive percentage change, problems related to examination, election, partial percentage and percentage error

**Income based questions** : concept of cost price, selling price and marked price, successive discount, problems based on number of articles and dishonest seller

### Unit III

**Flow chart** : problems based on data flow diagrams

**Alpha numeric coding** : coding and decoding, number series, alphabet series, alphanumeric series, alphabet test

### Unit IV

**Advanced ratio and proportion** : concept of ratio and proportion, problems based on ages and partnership

**Components and blending** : concept of alligation and mixtures, problems based on removal of mixture

### Unit V

**Counting methods** : principles of counting, numerical permutation(formation of numbers and sum of numbers), alpha permutation(rearrangement of words and rank of a word), circular arrangement, distribution based questions, formation of committee, geometry based problems

**Probability** : concept of probability, classification of events, conditional probability, problems based on coins dices and cards

### Unit VI

**Interest based problems** : concepts of interest calculations, difference between CI and SI, EMI calculations

**Analytical reasoning** : Direction Sense, Blood Relation

**Non verbal reasoning** : visual reasoning: mirror/water image, paper cutting and folding, completion of the figure, embedded figure, deviation of figure

### Text Books:

1. A MODERN APPROACH TO VERBAL AND NON-VERBAL REASONING by DR. R.S. AGGARWAL, S. CHAND & COMPANY
2. MAGICAL BOOK ON QUICKER MATHS by M.TYRA, BANKING SERVICE CHRONICLE

### References:

1. QUANTITATIVE APTITUDE FOR COMPETITIVE EXAMINATIONS by DR.R.S. AGGARWAL, S. CHAND & COMPANY

**References:**

2. MAGICAL BOOK SERIES ANALYTICAL REASONING by M.K. PANDEY, BANKING SERVICE CHRONICLE